

PortfolioIQ — Project Brief

CCSW 431 Agentic Development Workflow — Part Two, Lab 1: Project Selection & Agentic Foundation

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Problem Statement

Retail investors tracking a personal stock portfolio have no single, simple place to see what they own, how it's performing, and what it means for their risk and diversification. Spreadsheets show raw numbers but explain nothing; most portfolio apps never interpret them in plain language. PortfolioIQ closes that gap: an app that computes its own portfolio metrics and uses an LLM strictly to explain numbers it has already calculated — never to calculate them.

Target Users

Individual retail investors — particularly students and early-career professionals in Saudi Arabia managing a modest, self-directed stock portfolio — who want a fast, plain-language read on their holdings without needing financial-analyst background.

Core Features (Sprint 1 scope, this lab)

- Registration (Story 001) — email/password account creation via Firebase Authentication.
- Login & Authentication (Story 002) — email/password sign-in, one generic error message for both an unregistered email and a wrong password (test case TC006 — the exact issue security-reviewer found live in our Part One demo).

Core Features (later sprints, full project scope)

- Add/Edit Stock Holding; a Portfolio Dashboard computing position value, allocation %, P/L, P/L %, concentration %, and a risk score (Sprint 2).
- AI Portfolio Analysis and an AI Chat Assistant, grounded strictly in numbers the app has already computed — the LLM interprets, never calculates (Sprint 3).
- Settings & Profile (stretch, Sprint 4).

Intended Technology Stack

- **Client:** Android (Java, Android Studio) — not Kotlin.
- **Backend/auth/data:** Firebase Authentication + Cloud Firestore (no separate custom backend).
- **LLM:** Firebase AI Logic SDK calling Google Gemini directly from the app — satisfies both the course's Firebase and LLM requirements with one piece of infrastructure.
- **Live market data:** a free-tier stock API (Alpha Vantage / Finnhub / Twelve Data — final choice pending, does not block Sprint 1).
- **Architecture:** domain / data / presentation clean-architecture split (UseCase + Repository pattern), adapted from ECC's *android-clean-architecture* skill.
- **Agentic tool for this lab series:** Claude Code with the ECC plugin (*affaan-m/ECC*, MIT-licensed) — the same tool researched and demoed live in Part One, continued here per the brief's own guidance to "prefer the tool you are actually using in your build labs." Instructor-approved for Part Two.

Course: CCSW 431 Mobile Programming, University of Jeddah. This brief accompanies Lab 1's tool-setup evidence and first agent scaffolding session.